Project Initialization and Planning Phase

Date	27 June 2026
Team ID	SWTID-2026-9316
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform the credit card approval process using machine learning, improving prediction accuracy, efficiency, and transparency. It addresses the limitations of manual credit evaluation by providing an automated decision support system that reduces processing time, minimizes human error, and enhances customer satisfaction. Key features include a Random Forest-based prediction model, risk audit report generation, and real-time credit card approval prediction.

Project Overview	
Objective	The proposal report aims to transform the credit card approval process using machine learning, improving prediction accuracy, efficiency, and transparency. It addresses the limitations of manual credit evaluation by providing an automated decision support system that reduces processing time, minimizes human error, and enhances customer satisfaction. Key features include a Random Forest-based prediction model, risk audit report generation, and real-time credit card approval prediction.
Scope	The project collects applicant information, preprocesses the input data, performs feature engineering, predicts credit card approval using a Random Forest classifier, generates a detailed risk audit report, and stores prediction history for future reference and analysis.
Problem Statement	
Description	Financial institutions often rely on manual evaluation or traditional rulebased methods for credit card approval, resulting in delays, inconsistent decisions and increased operational effort. Applicants also experience uncertainty due to lengthy approval processes and limited transparency.

Impact	Solving this problem improves decision accuracy, reduces processing time, minimizes financial risk, enhances operational efficiency, and provides applicants with faster and more transparent approval decisions.
Proposed Solution	
Approach	Employ Machine Learning, Flask, Python, and data preprocessing techniques to analyze applicant information, perform feature engineering, predict credit card approval using a Random Forest classifier, and generate detailed risk audit reports.

Key Features	• Machine learning-based credit card approval prediction. • Automated data preprocessing and feature engineering. • Random Forest classification model. • Risk audit report generation. • Prediction history storage using JSON. • Responsive Flask web interface. • Docker-based deployment support.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computing Resources	Intel Core i5 Processor
Memory	RAM specifications	8 GB
Storage	Disk space for application, models, and logs	256 GB SSD
Software		
Frameworks	software	Python, Flask
Libraries	Additional Libraries	Scikit-learn, Pandas, NumPy, Joblib, Flask, Gunicorn
Development Environment	IDE	Visual Studio Code
Data		
Data	Source, size, format	Credit card applicant dataset, trained Random Forest model (.joblib), JSON prediction history (prediction_summary.json)